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Effect of Chordwise Slots on Aerodynamic Efficiency

Sidaard Gunasekaran[1], Tim Gerham[2]

Abstract

The balance of induced and parasite drag on a NACA 0012 wing was affected by allowing the freestream to pass through the wing at multiple spanwise locations. Three wing configurations, one with eight holes equally spaced, the second with eleven holes elliptically spaced (resembling lift distribution), and the third without any holes were tested at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT). The force-based experiment indicates similar lift coefficient for the three cases at lower angles of attack which is surprising given the loss of several stagnation points on wings with holes. However, the parasitic drag coefficient (obtained through momentum deficit profiles in the wake) of the 8-hole and 11-hole wings are greater than the baseline. But significant reduction in wingtip vortex azimuthal velocity, circulation, and turbulent intensity was observed in the 11-hole case when compared to the baseline indicating the effectiveness of the additional streamwise momentum through the holes. The reduction in the mentioned wingtip vortex properties in the 11-hole case was due to decrement in distance between the free shear layer (FSL) and the wingtip vortex core, which increased FSL-wingtip vortex interaction.

1. Introduction

The objective of the research is to determine the effect of freestream passing through the wing on the parasite and induced drag. The properties in the free shear layer (FSL) is assumed to be a surrogate for the parasite drag and the properties in the wingtip vortex is assumed to be a surrogate for the induced drag. By allowing the freestream to pass through the wing at certain locations along the span (especially at the wingtips), air mass and momentum is changed in the wake which alters the signature of the inboard FSL and the outboard wingtip vortex and the interaction between the wingtip vortex and the FSL.

1.1 Affecting Parasite Drag:

According to Newton's second law, assuming steady, incompressible, inviscid, two-dimensional flow with no body force, the momentum deficit in the free shear layer wake (which represents the loss of freestream momentum), translates into the drag force experienced by the wing. Numerous active and passive flow control techniques were studied through experiments and computational methods to affect the lift coefficient, skin friction drag coefficient, pressure drag coefficient (and induced drag in case of finite wings) of streamlined and bluff bodies.

Passive methods such as modifying the surface condition (Runyan & Steers 1980) (smoothness and waviness) and geometric shapes (Bushnell 1989) have been taken into consideration to reduce parasite drag. Alteration of the geometry helps to maneuver the pressure gradient (Schubauer & Spangenberg 1960), which delays turbulence (Smith 1981) and prevents separation over the airfoils' suction surface (Smith et al. 1981). All these methods affect the distribution of vorticity without changing the momentum or mass in the wing wake. But similar or better drag reduction can be obtained by intentionally targeting the momentum in the wake through active control techniques such as suction and blowing.

It has long been known that the aft blowing through the base of a bluff body may significantly reduce its pressure drag, both for 2-D and 3-D bodies (Bearman 1966, 1967, Wood 1967, Leal & Acrivos 1969, Lewis & Chapkis 1969, Tanner 1975, Gai & Kapoor 1981, Delaunay & Kaitsis 1999, Grosce & Meier 2001). The main mechanism responsible for reducing the pressure drag behind a bluff body is the alteration

[1] Assistant Professor, Department of Aerospace Engineering, University of Dayton, OH - 45469
[2] Undergraduate student, Department of Aerospace Engineering, University of Dayton, OH - 45469

of the amount and distribution of vorticity introduced into the wake. Such flow control techniques are employed in bullet-shaped objects where the base pressure is increased through the bleed holes which significantly reduces the base drag. The base bleed reduces the kinetic energy of the wake globally due to the presence of vorticity of different sign and by the reduced concentration of the vortical structures.

Over the course of several years, similar flow control techniques were used on streamlined objects such as wings to reduce parasite drag. Prandtl (Schlichting 1968) was the one of the first to employed boundary layer suction on a cylindrical surface to delay boundary layer separation. Huang et al. (2004) investigated the suction and blowing flow control approaches for the NACA 0012 airfoil. When the jet location and angle of attack were combined, perpendicular suction at the leading edge from 0.075 to 0.125 of the chord length, increased the lift coefficient better than other suction situations. It was also found that tangential blowing at downstream locations, around 0.371 to 0.8 of the chord length, leads to a maximum increase in the lift coefficient value. Sahu and Patnaik (2011) attached a rotating element in form of an actuator disc, which was embedded on the leading edge of the NACA 0012 airfoil, to inject momentum into the wake region to achieve high-performance airfoils that enable delayed stall conditions and achieve high lift-to-drag ratios. Beliganur and Raymond (2007) applied an evolutionary algorithm to optimize flow control. Results of their study showed that the use of two suction jets along with two blowing jets enhanced the lift-to-drag ratio for a NACA 0012 airfoil.

1.2 Affecting Induced Drag:

Suction and blowing are also used to reduce induced drag and affect the wingtip vortex characteristics. Several passive control techniques such as endplates of different sizes and shape, winglets, tip sails, etc have been used to reduce induced drag and minimize wingtip vortex strength. These techniques have also been implemented in full scale airplanes due to the ease of scalability. But active control techniques such as spanwise blowing was found to be not beneficial in terms of drag reduction. Mineck (1995) demonstrated that the addition of spanwise blowing at low Mach numbers resulted in smaller increments in the lift and smaller decrements in drag. Since the jet momentum is greater than the drag reduction and did not yield in significant augmentation of lift, Mineck (1995) concluded that the spanwise blowing at the wingtip is not a practical means of improving the aerodynamic efficiency of moderate aspect ratio wings at cruise Mach number of modern airliners.

But the active control techniques proved to be effective in changing the wingtip vortex characteristics at low Reynolds numbers. Simpson et al (2002) showed that the maximum tangential velocity of the wingtip vortex was reduced by the wingtip blowing. Margaris and Gursul (2010) employed wingtip blowing in the spanwise and in the transverse (normal to the span) direction and observed that the blowing from pressure side of the wing resulted in a diffused vortex. When blowing was performed from the upper surface of the wing, it produced multiple coherent vortices in the near wake. Even though the blowing is effective in reducing tip vortex strength, the implementation of such techniques in full scale airplane requires complex mechanism which adds weight to the airplane and is not easily scalable.

A recent study done by Gunasekaran and Altman (2017) showed that the FSL strongly influences the rollup of the wingtip vortices at lower angles of attack. The interaction of the FSL with the wingtip vortex effectively inhibits the wingtip vortex rollup process resulting in significant changes in the wingtip vortex azimuthal velocity especially in the core boundary. This research explores a passive method to increase the FSL-wingtip vortex interaction to reduce the strength of the wingtip vortex. By allowing the freestream to pass through the holes designed in the wing, the momentum deficit in the free shear is changed which in turn is hypothesized to change the properties in the wingtip vortex. Two different configurations of the wings with holes were tested. The first configuration has eight holes equally spaced across the span and the second configuration has eleven holes spaced elliptically across the span with more number of holes concentrated at the wingtip. The elliptical spacing of the holes was designed to mimic the lift distribution on a wing. Since the lift production is lower at the wingtips when compared to the wing root, and experience high downwash and drag comparatively, it can be used to manipulate the roll-up process of the wingtip vortex without compromising the total lift production.

2. Experimental Setup

2.1 Wind Tunnel

All the experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30" x 30" x 96") test sections. The test section is convertible from a closed jet configuration to an open jet configuration with the freestream range of 6.7m/s (20 ft/s) to 40m/s (140 ft/s) at a freestream turbulence intensity below 0.1% measured by hot-wire anemometer. The tunnel also can vary the freestream velocity profile at up to 5 Hz and over 50% velocity amplitude using a downstream shuttering system.

All the experiments mentioned in the paper were done in the open jet configuration where an inlet of 76.2 cm x 76.2 cm opens to a pressure sealed plenum. The effective length of the test section in the open jet configuration is 182cm (72"). A 137cm x 137cm (44" x 44") collector collects the expanded air on its return to the diffuser. A photo of the UD-LSWT open jet configuration is shown in Figure 1. The velocity variation for a given RPM of the wind tunnel fan is found using a Pitot tube connected to an Omega differential pressure transducer (Range: 0 – 6.9 kPa).

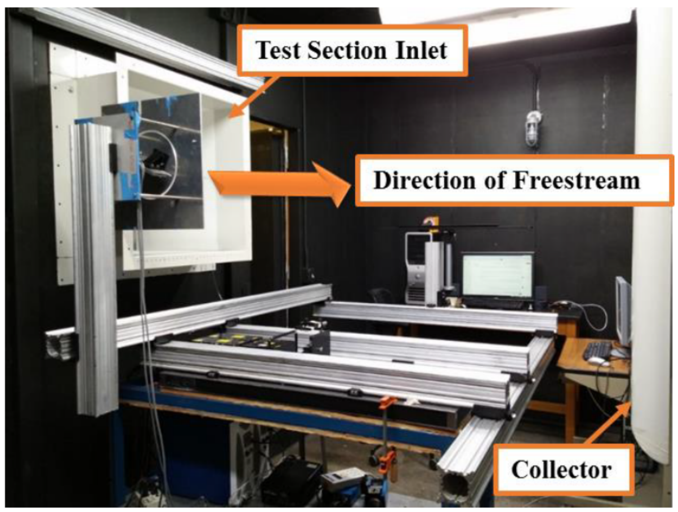

Figure 1 University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open-jet configuration

2.2 Test Models

Three configurations of NACA 0012 semi-span wing was modeled in SolidWorks: one with eight holes equally spaced along the span (Figure 2a), the second with eleven holes spaced elliptically along the span (Figure 2b) and the third with no holes for baseline study. The holes run along the entire chord length. The diameter of each hole (d_h) is 5 mm and the diameter to airfoil thickness ratio (d_h/t) is 0.41. The wings were then printed using Stratasys uPrint SE Plus printer at the University of Dayton. The 3D printed wings were carefully sanded using a 300 girt sand paper to reduce the surface roughness. The two configurations shown in Figure 2 can be differentiated by the net inlet momentum coefficient $C_{\mu_{inlet}}$ defined by,

$$C_{\mu_{inlet}} = n_{hole} * \frac{\dot{m}V_{hole\ inlet}}{q_\infty S} \tag{1}$$

where n_{hole} is the number of holes in the wing, $\dot{m}$ is the mass flow rate, $V_{hole\ inlet}$ is the velocity at the inlet of the hole, q_∞ is the dynamic pressure and S is the wing reference area. For the 8-hole wing, the net $C_{\mu_{inlet}}$ is 0.0152. For the 11-hole wing, the net $C_{\mu_{inlet}}$ is 0.020.

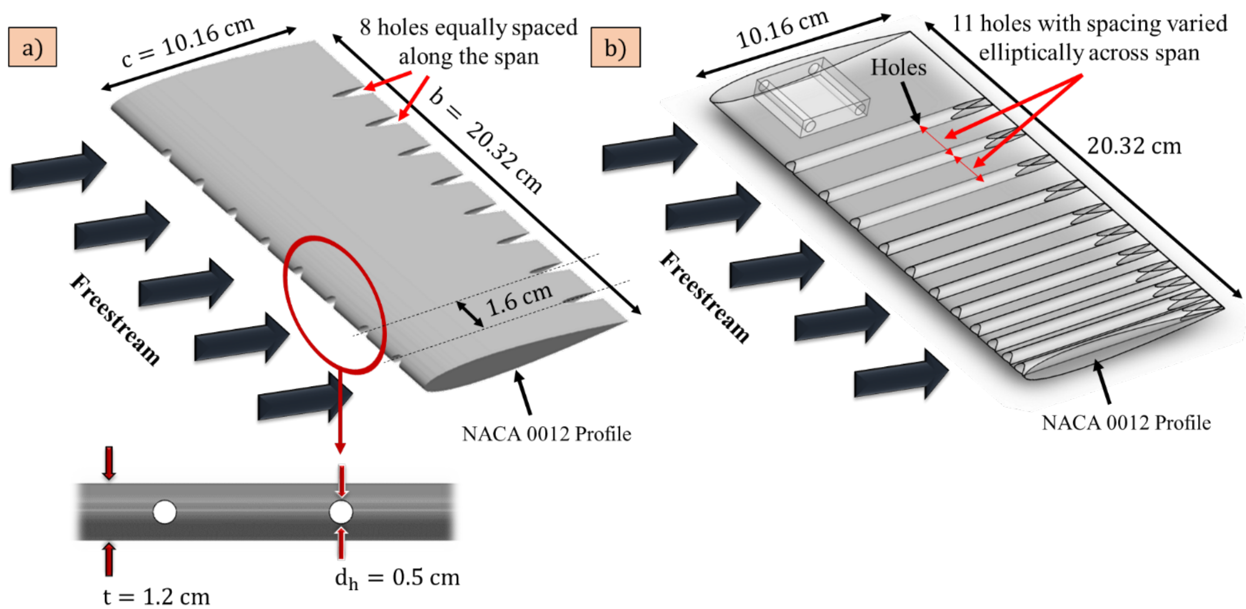

Figure 2 Schematic of a) NACA 0012 semi-span wing with eight holes equally spaced b) with eleven holes elliptically spaced

2.3 Force Based Experiment

The force-based experiment for all three cases was performed at a Reynolds number of 200,000 and an angle of attack range of -15 to 15 degrees. Two trials of the same experiment were done with increasing and decreasing the angle of attack to check for hysteresis. The test matrix of the force based experiment is shown in Table 1.

Table 1 Test Matrix for the force-based experiments

Airfoil	Semi-Span AR	Cases	Reynolds Number	Angle of Attack (Degrees)	Trials
NACA 0012	2	With 8 Holes Equally Spaces	200,000	-15 to 15	2
		With 11 Elliptically Spaced Holes			
		No Holes			

An ATI Mini-40 force transducer was secured underneath the wing at the quarter chord location which interfaced with the Griffin motion rotary stage to change the angle of attack. The rotary stage was controlled using the Galil motion software. The schematic of the test setup is shown in Figure 3. The root of the wing was made to be in alignment with the splitter plate. The freestream velocity was measured using a Pitot tube attached to an Omega differential pressure transducer (Range: 0-6.9 kPa).

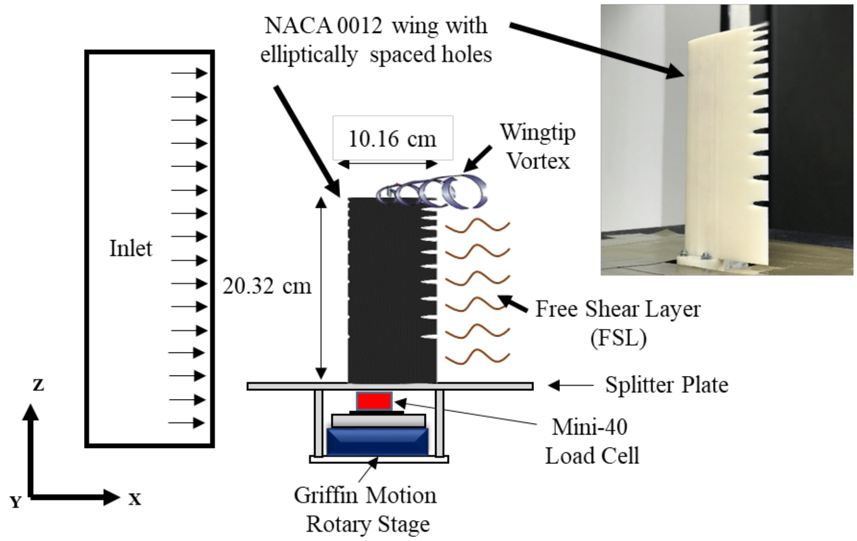

Figure 3 Schematic of the force based experiment test setup on a NACA 0012 wing with elliptically spaced holes (Dimensions are not to scale)

Force Transducer

An ATI Industrial Automation Mini-40 (www.ati-ia.com) sensor was used to measure the normal and axial forces. Lift and drag forces are obtained from the normal and axial forces using Equation 2 and 3.

$$L = N\cos\alpha - A\sin\alpha \tag{2}$$

$$D = N\sin\alpha + A\cos\alpha \tag{3}$$

The specifications for the Mini – 40 sensor are shown in Table 2. The normal and axial force was measured using the X and Y axis of the sensor with similar magnitude range. The resolution of the forces for all three axes is the same.

Table 2 Specifications of ATI Mini-40

	Fx (N)	Fy (N)	Fz (N)	Tx (Nm)	Ty (Nm)	Tz (Nm)
Range	40	40	120	2	2	2
Resolution	1/100	1/100	1/50	1/4000	1/4000	1/4000

The sampling rate during data acquisition from the Mini-40 was 100 Hz. Tare values were taken before and after each test, and then the average of the two tares are subtracted from the normal and axial force readings.

2.4 PIV Experiment

Streamwise Particle Image Velocimetry (PIV) and cross-stream PIV were conducted in the free shear layer and the wingtip vortex of the semispan wing models with and without the holes. The PIV measurements were obtained using a Vicount smoke seeder with glycerin oil and a 200 mJ/pulse Nd: YAG frequency doubled laser (Quantel Twins CFR 300). A Cooke Corporation PCO 1600 camera (1600 x 1200 pixel array) with a 105 mm Nikon lens was used to capture the images. One plano-convex lens and one plano-concave lens were used in series to convert the laser beam into a sheet. The laser and the camera were triggered simultaneously by a Quantum composer pulse generator. In each test case, over 700 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. A total of 2 iterations were performed during PIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. Both the streamwise and cross-stream PIV interrogations were conducted a Reynolds number of 135,000. The test matrix for the PIV experiment is shown in Table 3.

Table 3 Test Matrix for Free Shear Layer (FSL) and wingtip vortex interrogation using PIV

PIV Cases	Wing Model	Reynolds Number	Angle of Attack (Degrees)	Interrogation Location
Streamwise FSL	Baseline NACA 0012 wall-to-wall (2D)	135,000	0, 2, 4, 6, 8	Behind TE
	8-Hole Equally Spaced Wing wall-to-wall (2D)			Behind Hole
Cross-stream Wingtip Vortex	Baseline NACA 0012		2, 4, 6, 8	3 chord lengths downstream
	11-Hole Elliptically Spaced			

2.4.1 Streamwise PIV Setup:

The streamwise PIV was done in the FSL of the 8-hole wing with splitter plate on both wingtips to prevent wingtip vortex formation. The interrogation window was placed near the trailing edge of the wing as shown in Figure 4a. Two PIV interrogation regions were considered: one behind the trailing edge and the second behind one of the eight holes in the 8-hole wing. Nikon 105 mm lens was used in the streamwise PIV case which gave a spatial resolution of 292 pix/cm in both axes. The size of the field of view was 5.5 cm x 4.1 cm which gave a magnification factor of 0.21 (Figure 4b). The Δt for the images were set to obtain an average particle displacement of 8-10 pixels in the wake of the wing. The wing model was moved in the spanwise direction to perform PIV behind the hole to maintain the same magnification factor and resolution.

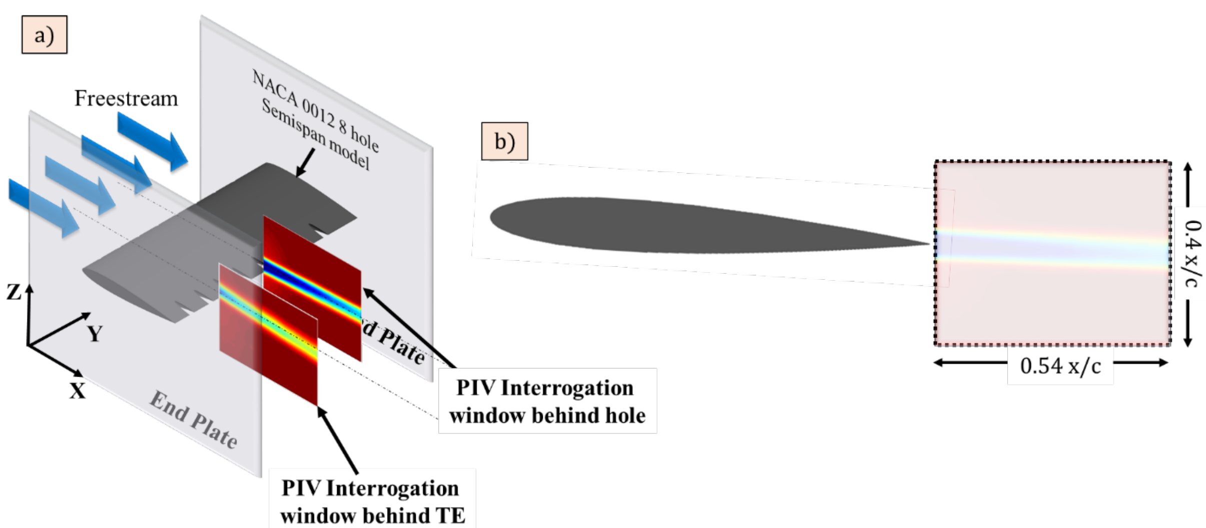

Figure 4 a) Schematic of the PIV interrogation windows in the wake of NACA 0012 wall-to-wall model b) Schematic of the PIV field of view in the wake of wing.

2.4.2 Cross-stream PIV setup:

The cross-stream PIV was done to determine the effect of added streamwise momentum from the holes on the roll-up of the wingtip vortex. The wingtip vortex from the baseline NACA 0012 and the 11-hole elliptically spaced wing were interrogated at multiple angles of attack at a Reynolds number of 135,000 as mentioned in Table 3. The schematic of the cross-stream PIV setup is shown in Figure 5. The cross-stream interrogation window is 3 chord lengths downstream from the trailing edge of the wing. The PCO 1600 camera was located at more than ten chord lengths downstream from the trailing edge of the wing, inside the collector so the effect of the camera being in the flow would comparatively be less. Due to the long object distance, a Nikon 200 mm with extension rings were used to focus the interrogation region. The spatial resolution of the field of view was 350 pix/cm, and the size of the field of view was around 4.6 cm x 3.4 cm which gave a magnification factor of 0.25. The time delay between the laser pulses was changed at each angle of attack to obtain 8-10 particle displacement at the boundary of the wingtip vortex.

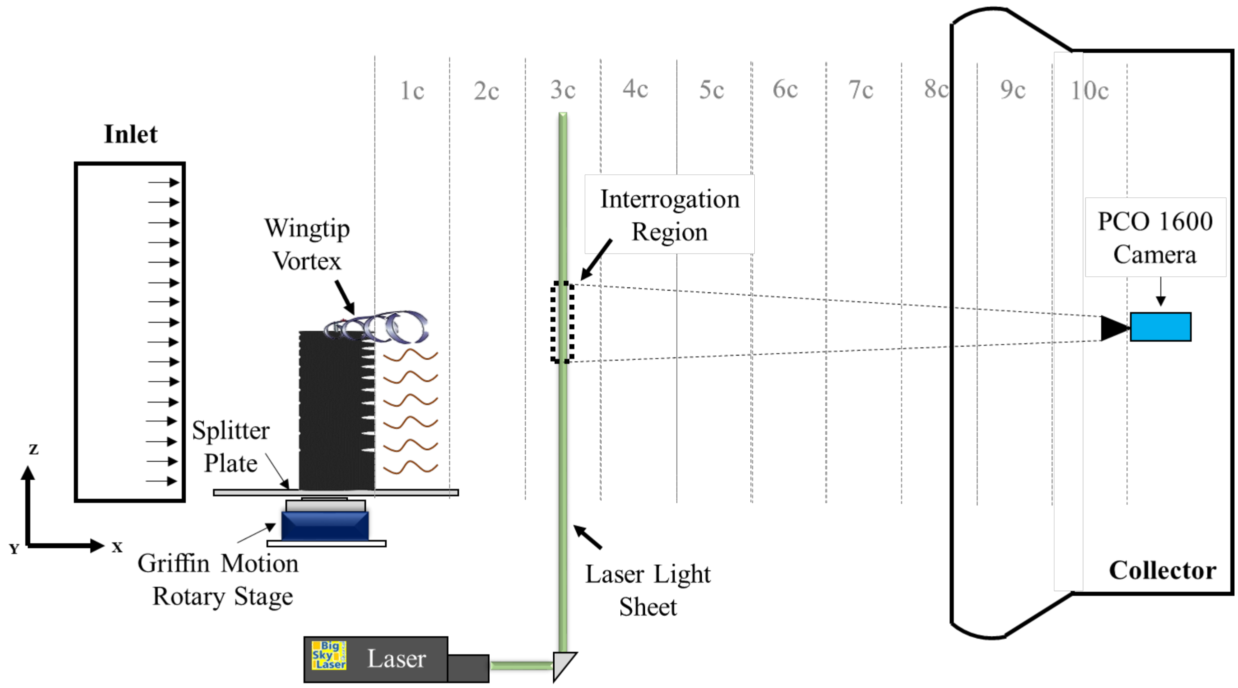

Figure 5 Schematic of cross-stream PIV setup for wingtip vortex interrogation.

3. Results

3.1 Force-Based Experiment Results:

The results from the force based experiments for Reynolds number of 200,000 are shown in Figure 6. According to McCormick's formula (McCormick (1995)), the lift curve slope depends on the aspect ratio by,

$$a = \frac{dC_L}{d\alpha} = a_0 \left(\frac{AR}{AR + 2\frac{(AR+4)}{(AR+2)}} \right) \tag{4}$$

where $a_0 = 2\pi$, according to thin airfoil theory. The best fit line of the lift curve gives an effective aspect ratio of 2 which is smaller than the intended aspect ratio of 4. The reduction in effective aspect ratio could be due to the wing-splitter plate interface contributing to three dimensionality of the flow. The baseline results show a good match with the results from Ngo and Barlow (2002) for a Reynolds number of 480,000 for AR 2. The comparison of lift coefficient magnitude between the baseline and the 8-hole wing ($C_{\mu_{inlet}} = 0.015$) shows almost no variations as a function of angle of attack (Figure 6a). Any changes in lift coefficient falls between the uncertainty band of the sensor as indicated by the error bars. Similar observations can be seen for the 11-hole wing. Even with significant air mass going through the wing and with the loss of stagnation points in the leading edge, similar lift coefficient magnitude is seen between the baseline and the 11-hole wing. Any differences in the lift at lower angles of attack between the baseline and the 11-hole wing fall within the uncertainty band of the sensor as indicated by the error bars. Measurable reduction in lift is only in the order of 12% at 14° angle of attack which was surprising. The implications of this result are that the weight of the wing can be reduced without sacrificing the wing performance in lift. Assuming the wing models is made of aluminum ($\rho_{Aluminum} = 2700\ kg/m^3$), the differences in the weight of the wing between the baseline wing and the 11-hole wing is around 13%. This means that at lower angles of attack, the wing could achieve the same lift with less weight resulting in reduced wing loading which leads to better takeoff, climb and turn performance.

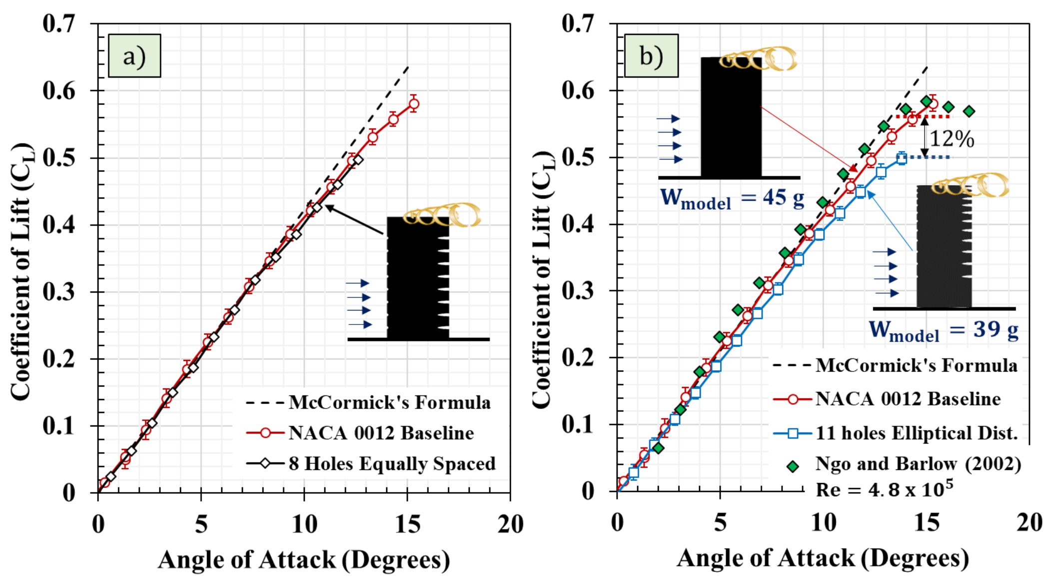

Figure 6 Variation of coefficient of lift for a) NACA 0012 wing with eight holes equally spaced b) NACA 0012 wing with eleven holes elliptically spaced

The reduction in the lift at higher angles of attack could be due to the removal of stagnation points at multiple hole locations or the disruption of wingtip vortex due to the streamwise momentum of air going through the wing at the wingtip location or both. The differences in lift is used to calculate the differences in the induced drag. The induced drag was found by,

$$C_{D_{Induced}} = \frac{C_L^2}{\pi e AR} \tag{5}$$

where e is the span efficiency and AR is the aspect ratio. The span efficiency of the baseline and the 11-hole wing was found using the lift curve slope equation from thin airfoil theory (Equation 6).

$$a = \frac{a_0}{1 + \frac{a_0}{\pi\ e\ AR}} \tag{6}$$

Where a is the lift curve slope of the finite wing and $a_0 = 2\pi$. From Equation 6, the span efficiency for the baseline was 0.69 and the span efficiency for the 11-hole wing was 0.65. Keeping in trend with the lift variation, the magnitude of induced drag is decreased in the 11-hole wing case when compared to the baseline. The differences in the induced drag between the baseline wing and the 11-hole wing increases as a function of angle of attack (Figure 7). A maximum induced drag reduction of 20% is observed at an angle of attack of 14°. The ATI mini-40 sensor could not resolve the magnitude of the total drag forces of the wing at lower angles of attack. Any differences in the drag forces between the baseline wing and the 11-hole elliptically spaced wing lie within in the uncertainty band of the sensor. Therefore, Particle Image Velocimetry (PIV) was used to determine the wake momentum deficit and the parasitic drag of the wing configurations. The results from the PIV are discussed in the section below.

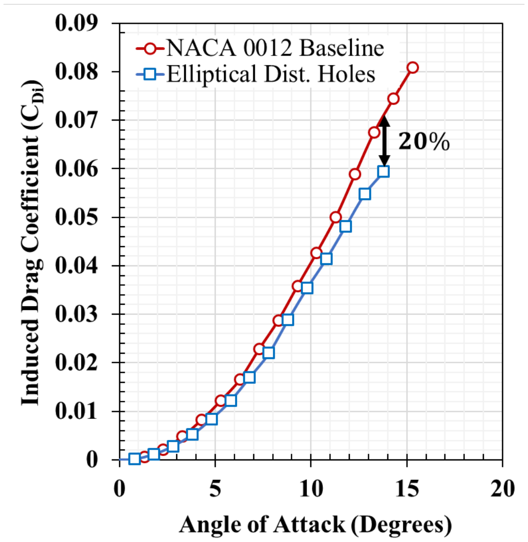

Figure 7 Change in induced drag as a function of the angle of attack for baseline wing and 11-hole elliptically spaced wing. The 11-hole wing creates less induced drag than the baseline.

3.2 Streamwise PIV results

The streamwise velocity U_x contour obtained behind the hole and behind the trailing edge of the 8-hole wing is shown in Figure 8a. A higher momentum deficit behind the hole was observed in all angles of attack when compared to the trailing edge wake. The higher deficit is due to the boundary layer growth inside the hole. However, the differences in the momentum deficit between the two cases decreases with increase in angle of attack. At 8°, both cases show a similar momentum deficit. These observations are more apparent in the momentum deficit profiles shown in Figure 8b. The profiles were taken by averaging 10 data columns in the center of the field of view. The normalized streamwise velocity is plotted against the normalized wake-half width z/L_0 where L_0 is the wake half-width which is considered as the location of 99% U_∞. The profiles indicate that the momentum deficit behind the hole is higher than the momentum deficit behind the trailing edge at all angles of attack. The peak normalized axial velocity behind the hole at 4° has the same magnitude as the peak normalized axial velocity behind the trailing edge at 8° angle of attack. This indicates a significant increase in parasite drag behind the hole. The momentum deficit profiles shown in Figure 8b were used to determine the total parasitic drag coefficient of the baseline NACA 0012 wing, the 8-hole equally spaced wing and the 11-hole elliptically spaced wing. The total parasitic drag coefficient of the wing was found by integrating the sectional drag coefficient along with wingspan.

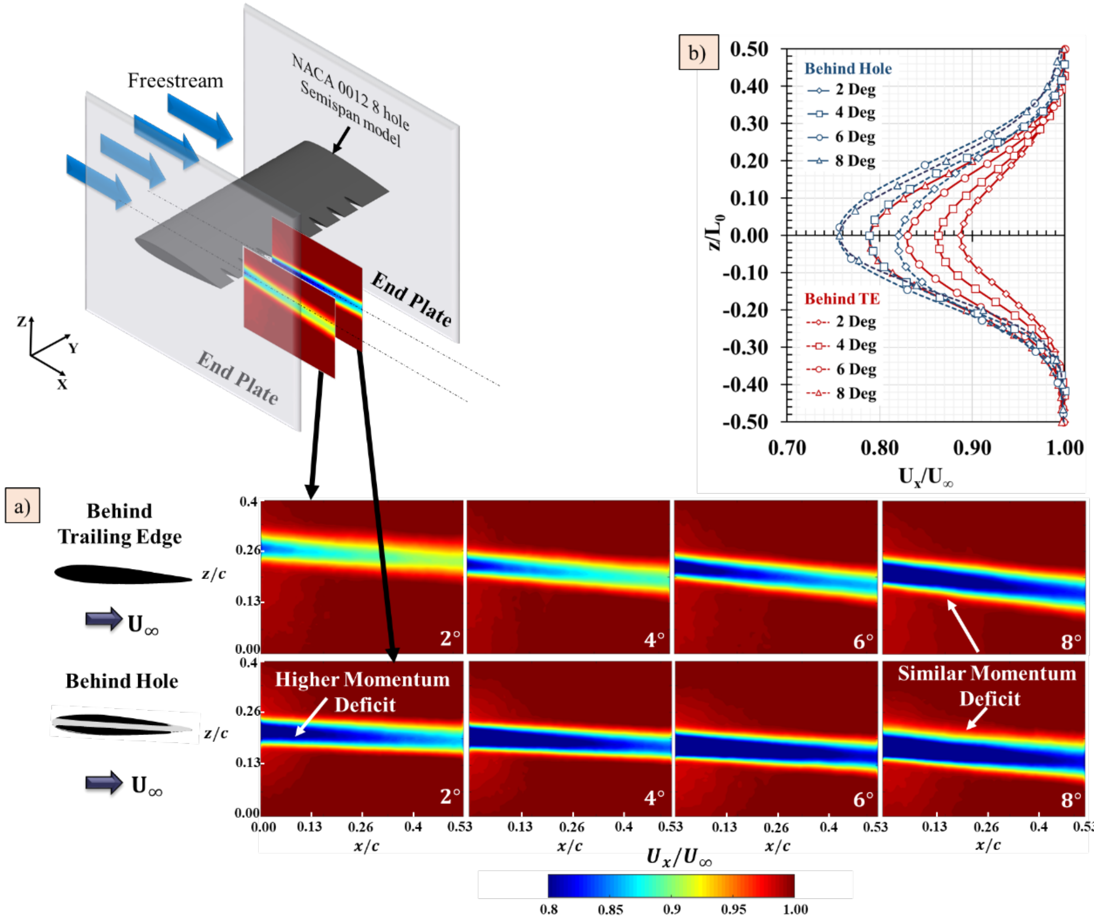

Figure 8 a) Streamwise velocity contours in FSL behind the trailing edge of the NACA 0012 wing and behind the hole b)Variation of the momentum deficit behind the trailing edge of the wing and the hole. The momentum deficit in the wake of the hole is larger than the momentum deficit in the wake behind the trailing edge.

The sectional drag coefficient behind the trailing edge and the hole was determined using Equation 7.

$$C_D = \frac{\rho U_\infty^2}{q_\infty c} \int_{-\infty}^{\infty} \frac{U_x}{U_\infty}\left(1 - \frac{U_x}{U_\infty}\right) dz \tag{7}$$

where ρ is the air density, q_∞ is the dynamic pressure and c is the chord length of the airfoil. The drag coefficient variation with angle of attack is shown in Figure 9. As expected, the drag coefficient behind the trailing edge varies non-linearly with the angle of attack, and it is interesting to note that the drag coefficient varies almost linearly with the angle of attack behind the hole. The magnitude of drag coefficient behind the hole is significantly higher at lower angles of attack than the drag coefficient of the rest of the span where there are no holes. At 8° angle of attack, the sectional drag coefficient behind the hole and the trailing edge is almost similar. To determine the total parasitic drag coefficient of the entire wing with and without holes, a theoretical distribution of the sectional drag coefficient is plotted in Figure 10a for 8-hole wing using the drag coefficient values obtained from the momentum integral. The sections of the wing with NACA 0012 profile has a lower drag coefficient than the wing sections with holes as observed in Figure 9. The theoretical drag coefficient distribution increases as the angle of attack increases. The total drag coefficient of the wing was found by integrating the sectional drag coefficient along the span of the wing. The net parasitic drag coefficient of the NACA 0012 baseline wing, the 8-hole wing, and the 11-hole wing is shown in Figure 10b for multiple angles of attack. Both the 8 and 11-hole wing shows a measurable increase in drag coefficient when compared to the baseline. But the differences in the drag coefficient between the 8-hole and 11-hole wing is almost negligible. This result implies that the 11-hole wing reduces induced drag (Figure 7) at higher angles of attack without exhibiting a significant increase in the parasitic drag. The 11-hole elliptically spaced wing also has an added advantage of reduced weight (Figure 6b) which aids aircraft performance.

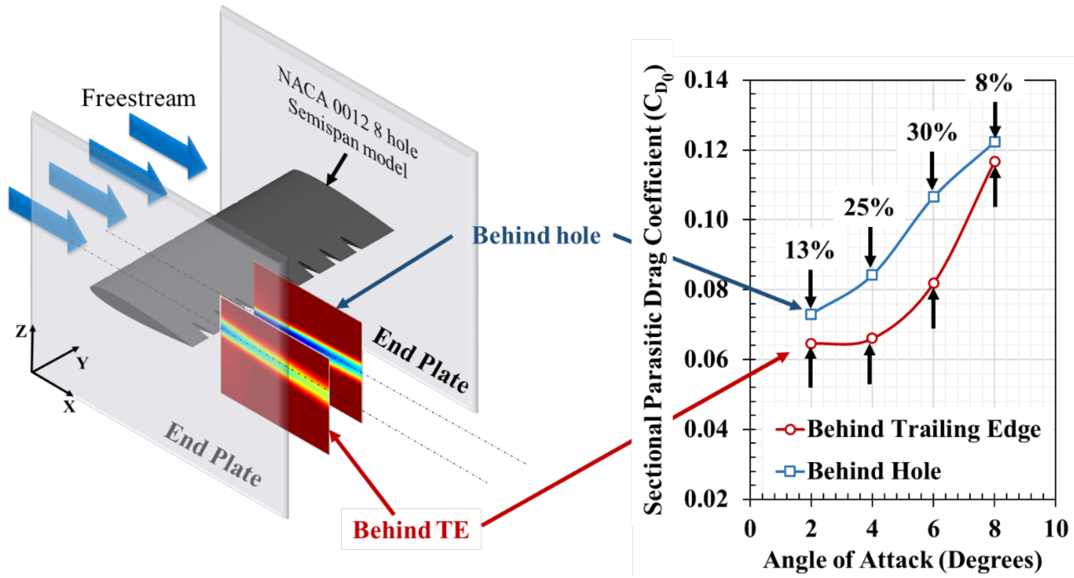

Figure 9 Variation of coefficient drag estimation behind the trailing edge and the hole as a function of the angle of attack. The estimated drag coefficient behind the hole is higher than the drag coefficient behind the hole.

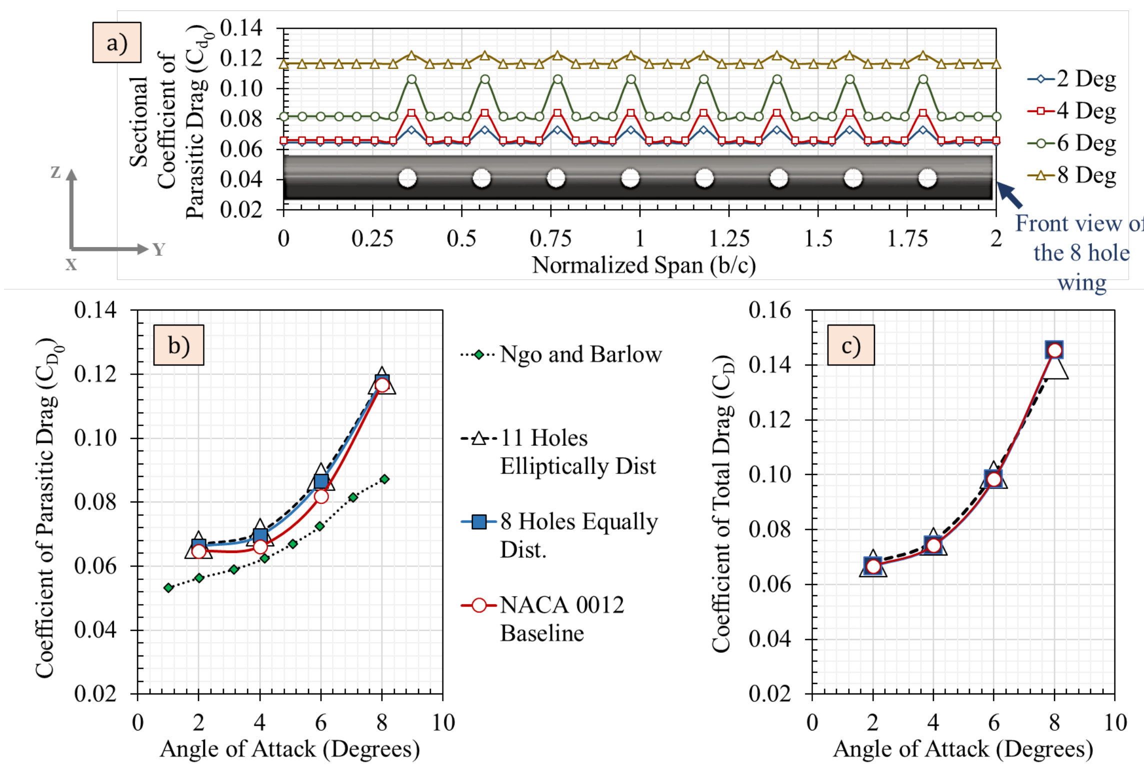

Figure 10 a) Section drag coefficient variation across the span for 8 hole wing. The drag coefficient behind the hole is higher than the drag coefficient behind the trailing edge of NACA 0012. b) The variation of net parasitic drag coefficient and c) total drag coefficient of the baseline NACA 0012, eight holes equally spaced wing and 11-hole elliptically spaced wing.

Adding the induced drag found from force based experiment and parasitic drag data found from PIV, the total drag coefficient for the baseline and the 11-hole wing is shown in Figure 10c. The total drag for all three wing configurations remains the same which is interesting. The reduction in induced drag in the 11-hole wing case is compensated by the increase in the parasite drag. Total drag reduction in the order of 4% is observed at an angle of attack of 8° in the 11-hole wing. Even though the total drag reductions using the 11-hole wing is only by a few percent, the added streamwise momentum in the wake significantly impacted the formation and rollup of the wingtip vortex which will be discussed below. Including the added benefit of weight reduction by 13% while maintaining the same aerodynamic performance makes the 11-hole elliptically space wing configuration desirable for airplane wings, helicopter blades, propellers and turbines.

3.3 Wingtip vortex PIV results:

3.3.1 Vortex Wandering:

The oscillations of the wingtip vortex in the cross-stream plane is typically referred to as vortex wandering or meandering. The wandering biases the vortex measurements and cause it to appear with large vortex radius and with less vorticity and circulation than the actual ones. Therefore, the vortex wandering was quantified by tracking the vortex center across all image pairs. The resultant deviations of the wingtip vortex center for both the baseline NACA 0012 and 11-hole wing case are shown in Figure 11a at 6° angle of attack. The tip vortex wandering can be quantified by the magnitude of RMS wandering amplitude σ (Heyes et al (2004)) given by,

$$\sigma = \sqrt{\sigma_y^2 + \sigma_z^2} \tag{8}$$

Where σ_y is the RMS wandering amplitude component in y-direction and σ_z is the RMS wandering amplitude component in the z-direction. From Figure 11a, the RMS wandering amplitude for both the baseline and the 11-hole wing was found to be 34 mm.

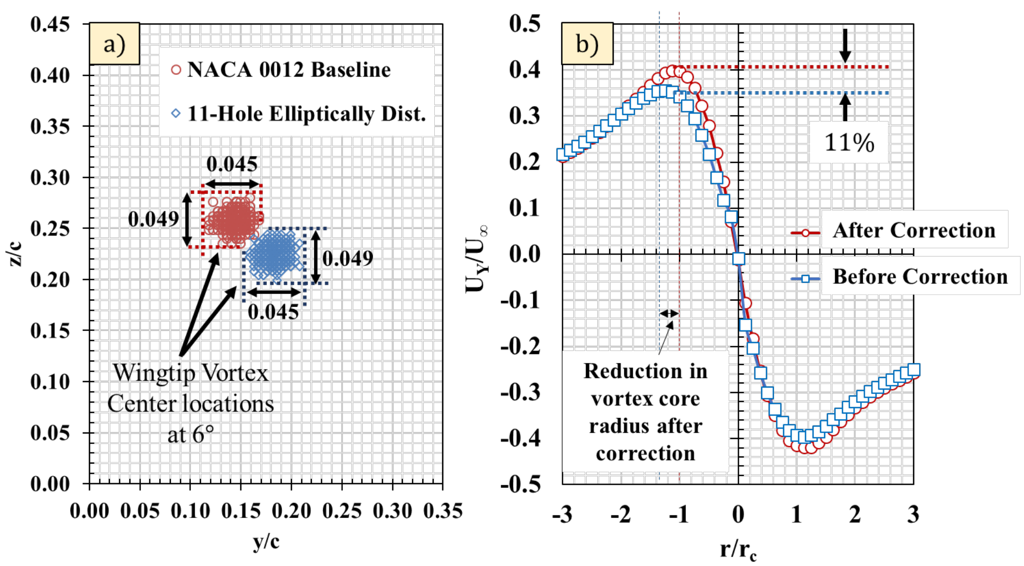

Figure 11 a) Wingtip vortex wandering for NACA 0012 baseline and 11-hole wing at 6° angle of attack b) Wandering corrected azimuthal velocity profile of baseline wing at 8° angle of attack.

Similar wandering was observed for 4° and 8° cases with the wandering magnitude less than 5% of the chord. Giacomo (2017) postulated that the circulation is main parameter controlling the vortex wandering and the magnitude of wandering reduces with increase in angle of attack and reduced downstream distances. Since the PIV were taken at three chord lengths downstream, the wandering of the vortex is contained within 5% and will only decrease with increase in angle of attack. Identifying the vortex center precisely at 2° angle of attack proved to be a challenge due to a much smaller and weaker vortex. Therefore, only the image pairs which showed vortex center deviation within 5% were taken into account for the 2° angle of attack case. The wandering corrected azimuthal profile is shown in Figure 11b for the baseline wing at 8° angle of attack. As suspected, there was a 11% increase in peak tangential velocity and reduction in the vortex core radius in the wandering corrected case when compared with no wandering correction.

3.3.2 Identifying Vortex Center Location

The location of the wingtip vortex center was determined by the scaled Q-criterion. The Q-criterion is a well-established method to find vortices in the flow. The equation for Q-criterion is,

$$Q = \frac{1}{2}\left(||\Omega||^2 - ||R||^2\right) \tag{9}$$

Where $||\Omega||$ is the absolute magnitude of vorticity given by

$$\Omega = \frac{1}{2}\left(\frac{\partial U_z}{\partial y} - \frac{\partial U_y}{\partial z}\right) \tag{10}$$

and $||R||$ is the absolute magnitude of strain rate given by

$$R = \frac{1}{2}\left(\frac{\partial U_z}{\partial y} + \frac{\partial U_y}{\partial z}\right) \tag{11}$$

The scaled Q-criterion is the shear strain normalized Q-criterion,

$$Q_s = \frac{1}{2}\left(\frac{||\Omega||^2}{||R||^2} - 1\right) \tag{12}$$

Since the magnitude of vorticity is larger at the vortex center and the magnitude of shear strain is smaller at the vortex center, the Q-criterion distinctly identifies the vortex center by having a strong peak at vortex center location. An example plot of the normalized Q-criterion of the 11-hole wing case at 8° angle of attack is shown in Figure 12. The maximum Q-criterion was identified as the center of the wingtip vortex.

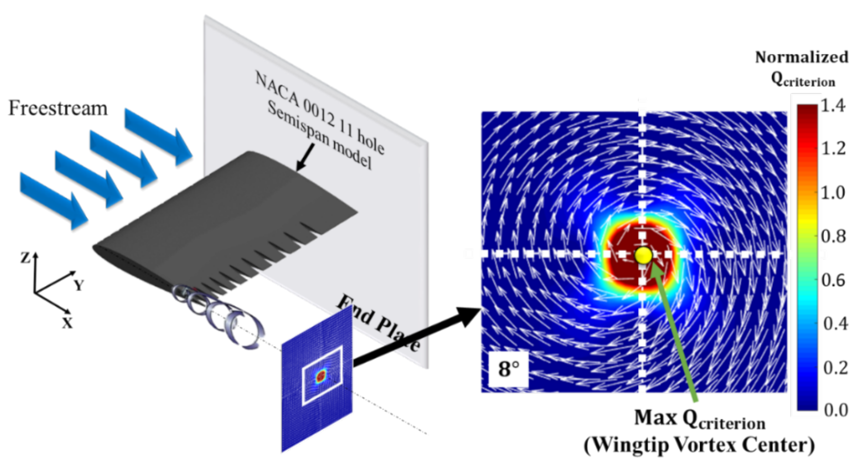

Figure 12 Normalized Q-criterion for wingtip vortex at 8° angle of attack behind 11-hole elliptically spaced wing

3.3.3 Wingtip Vortex Azimuthal Velocity

The 11-hole wing configurations showed reductions in induced drag (Figure 7) which could occur due to changes in wingtip vortex properties. To test this hypothesis, the cross-stream PIV results are analyzed for the baseline and the 11-hole wing configuration in this section. The azimuthal velocity contour (U_z) is shown in Figure 13 for 2°, 4°, 6° and 8° angle of attack for both the baseline NACA 0012 wing and the 11-hole elliptically spaced wing. The magnitude of azimuthal velocity increases with increase in angle of attack for both cases but the differences in the magnitude of the azimuthal velocity can be clearly seen across all angles of attack. The contours show distinct and significant reduction in the magnitude of the wingtip vortex azimuthal behind the 11-hole wing when compared to the baseline. This reduction in azimuthal velocity magnitude could be due to the streamwise momentum through the hole impinging the wingtip vortex. Similar results were observed for the U_Y azimuthal velocity as well. To quantify the decrement in the peak tangential velocity in the 11-hole wing case, the tangential velocity profiles were taken across the wingtip vortex center. Vertical (Z-direction) and horizontal (Y-direction) profiles were taken across the vortex center to determine azimuthal velocity profiles. The profiles were taken at the wingtip vortex center location.

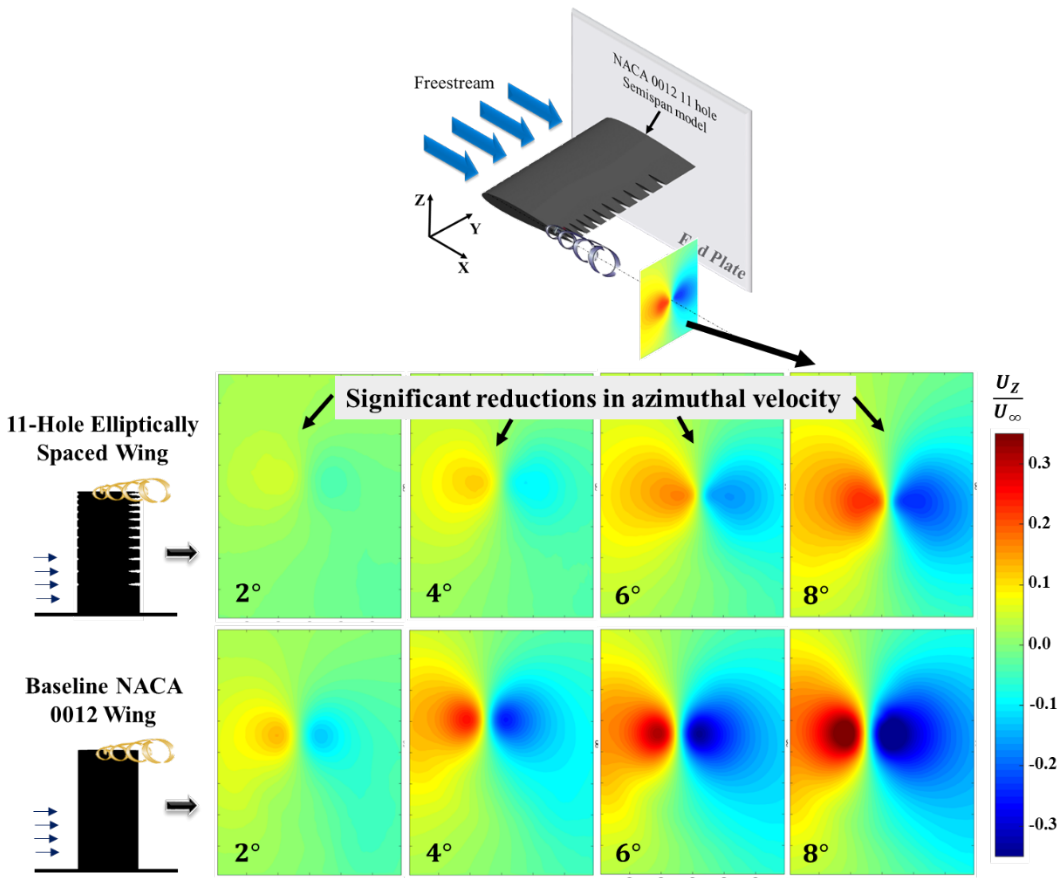

Figure 13 Variation of azimuthal velocity U_z for a) 11-hole elliptically spaced wing b) baseline NACA 0012 wing for different angles of attack. The magnitude of the azimuthal velocity of the 11-hole wing case is significantly lower than the baseline for all angles of attack.

The azimuthal velocity profiles are shown in Figure 14a for the baseline wing and the 11-hole wing. The azimuthal velocity profiles show a dramatic decrease in the magnitude when compared to the baseline NACA 0012 wing. The azimuthal velocity at 8° angle of attack in the 11-hole wing case has the same magnitude as the azimuthal velocity at 4° angle of attack in the baseline case. This trend shows an opposite behavior of what is seen in the momentum deficit profiles (Figure 8b). The profiles shown in Figure 14a is taken in the spanwise (Y-direction) plane across the wingtip vortex. Similar magnitudes and trends were observed in the vertical plane (Z-direction) as well. The peak azimuthal velocity is shown for both the cases in Figure 14b. The variation of peak azimuthal velocity is linear in both cases as shown by the linear regression line with the baseline wing having the higher magnitude. The differences between the peak azimuthal velocity between both cases decreases with increase in angle of attack from 67% at 2° angle of attack to 43% at 8° angle of attack. The average differences in the peak azimuthal velocity between the both cases are more than 50%. Interesting comparisons can be made between the results shown in Figure 14 and the results from Gunasekaran and Altman (2017) who quantified the changes in the wingtip vortex due to the presence of spanwise boundary layer trip on a AR 4 flat plate. While the spanwise boundary layer trip increased the momentum deficit in the free shear layer, it was also found to increase the peak azimuthal velocity of the wingtip vortex. But Figure 14 shows an opposite behavior. While the freestream passing through the holes increased the momentum deficit in the FSL (just like the spanwise boundary layer did in Gunasekaran and Altman (2017)), it significantly decreased the peak azimuthal velocity in the wingtip vortex. The turbulent parameters such as the normalized root-mean-square values will be discussed later in the paper to get more insight into the relationship between the FSL and the wingtip vortex. Figure 14 clearly shows that the magnitudes of the azimuthal velocity profiles are affected by the changes in the streamwise momentum.

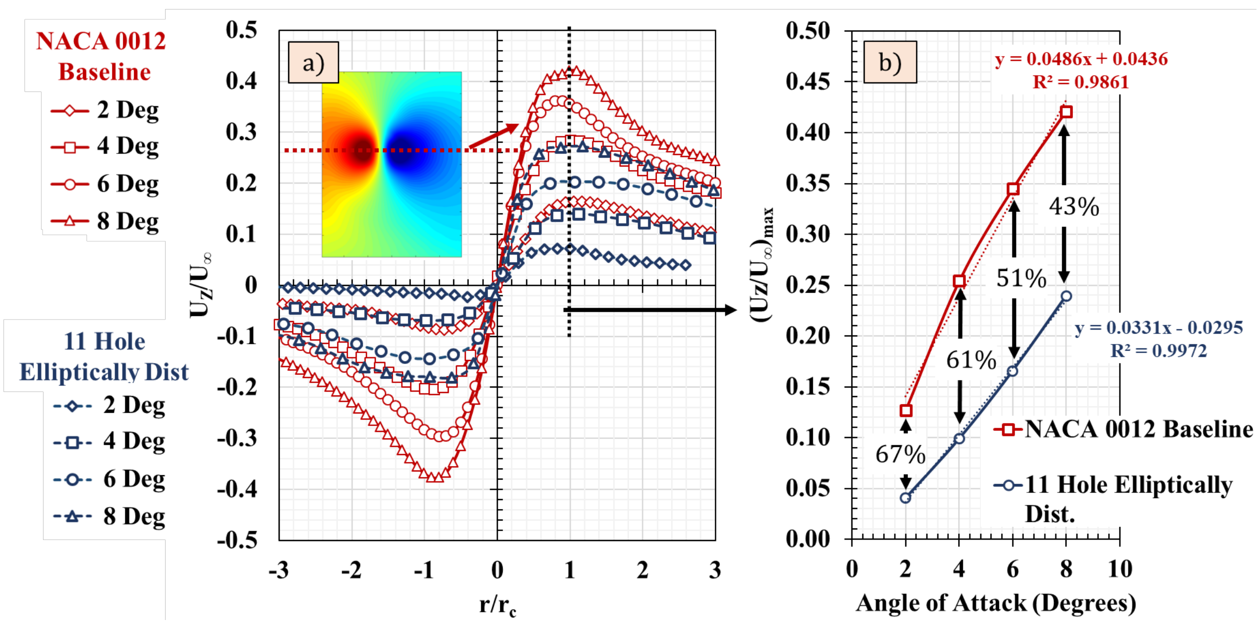

Figure 14 a) Azimuthal velocity variation as a function of normalized radius for different angles of attack. The magnitudes of the azimuthal velocity dramatically decreased when compared to the NACA 0012 baseline wing. b) The peak azimuthal velocity variation as a function of the angle of attack shows an average of 50% decrease between the baseline and 11-hole elliptically spaced wing.

To quantify the variations in the trends of the azimuthal velocity, a normalized azimuthal velocity is compared against the well-known Batchelor's model (Batchelor 1964) for the azimuthal velocity variation. Batchelor took a non-viscous (inviscid) approach by assuming that the pressure in the core of the vortex changes as it convects downstream. Batchelor vortex model for the distribution of azimuthal velocity is shown in Equation 13.

$$\frac{U_Z}{U_{Z_{max}}} = \left(1 + \frac{1}{2\alpha}\right)\frac{1}{\eta}\ (1 - \exp(-\alpha\eta^2)) \tag{13}$$

where η is the normalized vortex radius r/r_c with r_c being the core radius and α is Lamb's constant with a value of 1.256. The experimental azimuthal velocity profiles with Batchelor's model is compared to show the variation in the trends of the azimuthal velocity profiles between the baseline and the 11-hole wing. The azimuthal velocity profiles were taken horizontally (Y-direction) and vertically (Z-direction) across the wingtip vortex to determine the changes in the trend due to the proximity of the FSL. The horizontal profile of U_z contour is taken where the free shear layer rolls up into the wingtip vortex (can be seen clearly in the vorticity plot shown in Figure 17). The vertical profile of U_Y contour is taken in the area away from the free shear layer as indicated by the dotted lines in the contour shown in Figure 15. Both the horizontal and vertical azimuthal profiles show reasonably good match in the core of the wingtip vortex with Batchelor's model. Deviations are seen in the vortex boundary in both the vertical and horizontal profiles. The normalized U_z profile, being close to the free shear layer shows higher deviations from the Batchelor's model than the U_Y profile. The higher angle of attack cases (6° and 8°) matches extremely well with the Batchelor's model. At lower angles of attack, the deviations from the Batchelor's model are due to the proximity of the FSL to the wingtip vortex and increased FSL-wingtip vortex interaction. This observation is made in Gunasekaran and Altman (2017) as well.

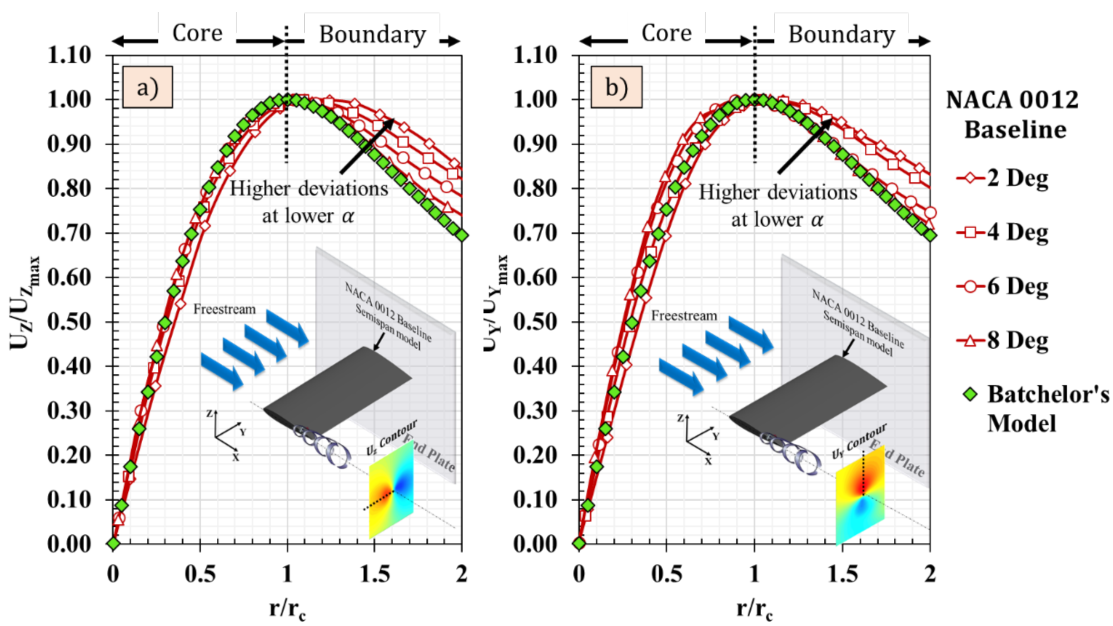

Figure 15 a) Uz azimuthal velocity profile (near FSL) and b) Uy azimuthal velocity (away from FSL) of NACA 0012 baseline wingtip vortex is compared with Batchelor's model. The experimental velocity profiles agree well with Batchelor's model in the core and deviate from the Batchelor's model in the boundary. This deviation could be due to the free shear layer interaction with the wingtip vortex.

On the other hand, the azimuthal velocity profiles of the 11-hole elliptically spaced wing show significant and larger deviations from the Batchelor's model at the wingtip vortex core and the boundary (Figure 16). Both the U_z and U_Y profiles show deviations in the wingtip vortex core from the Batchelor's model suggesting an increased FSL-wingtip vortex interaction. In the vortex core the profiles deviate from the Batchelor's model by an average of 10% and the deviations in the vortex core boundary from the Batchelor's model are in the order of 30% or greater. These are significant deviations which indicates that the wingtip vortex is severely affected by the streamwise momentum coming through the holes. Such deviations could occur if the FSL interacted strongly with the rollup of the wingtip vortex.

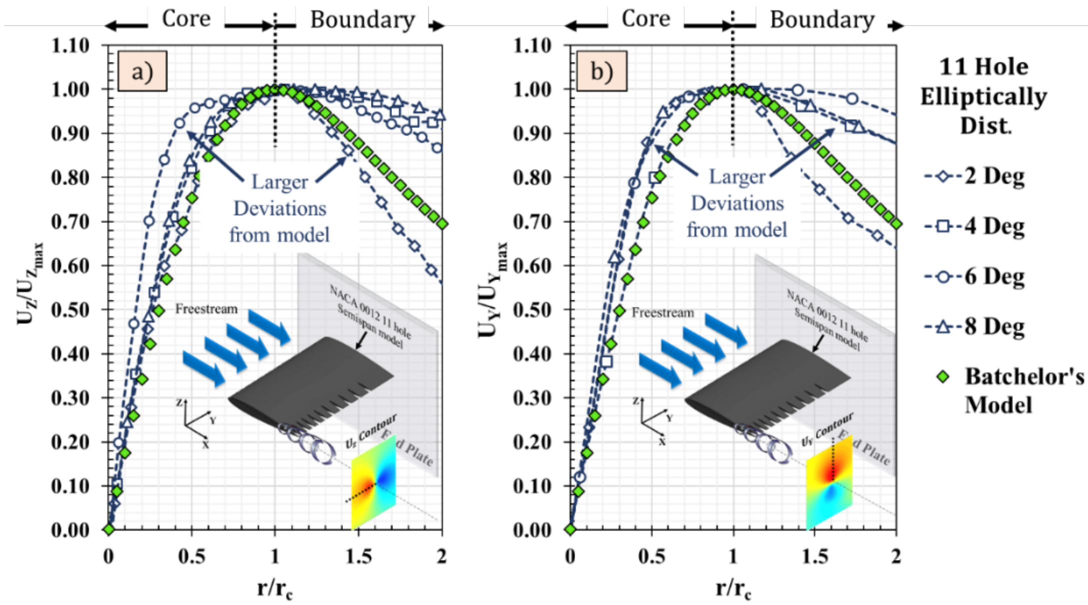

Figure 16 a) Uz azimuthal velocity profile (near FSL) and b) Uy azimuthal velocity (away from FSL) of 11-hole distributed wing is compared with Batchelor's model. Significant deviations from the model are observed at both the wingtip vortex core and the boundary.

3.3.4 Wingtip Vortex Vorticity and Circulation

The wingtip vortex vorticity was determined to get an insight into the FSL interaction by observing the distance between the wingtip vortex core and the FSL. The normalized x-vorticity of the wingtip vortex was determined using Equation 9 for both the baseline and the 11-hole wing cases (Figure 17).

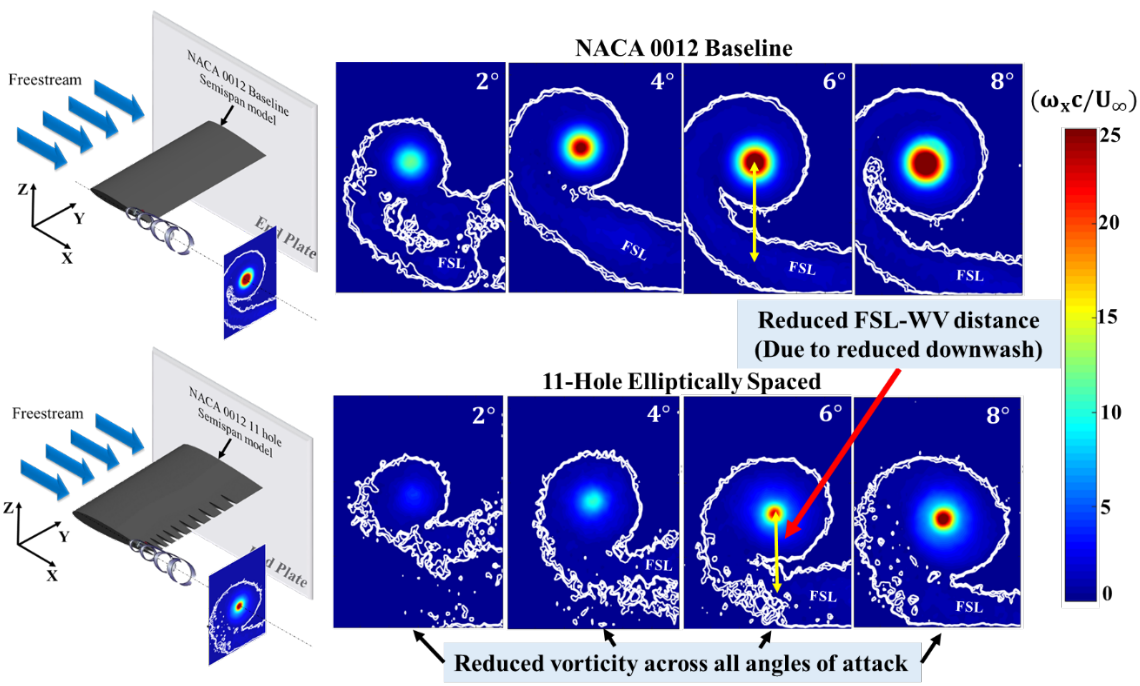

Figure 17 Normalized vorticity variation across different angles of attack for baseline NACA 0012 wing and 11-hole elliptically spaced wing. Significant reductions in vorticity can be observed in the 11-hole wing case.

The magnitude of vorticity increases with increase in angle of attack for both cases as expected. But the maximum normalized x-vorticity at the core of the wingtip vortex is significantly lower in the 11-hole wing case when compared to the baseline. This provides more evidence that the added streamwise momentum in the wake due to air passing through the holes strongly influenced the roll-up of the wingtip vortices. The normalized x-vorticity values of 0.15, 0.20 and 0.30 are highlighted in white to indicate the presence of the free shear layer (FSL) in the cross-stream plane.

The FSL plays a significant role in the development process of the wingtip vortex. It is postulated through evidence in Gunasekaran and Altman (2017) that the free shear layer moves away from the wingtip vortex in the vicinity of maximum aerodynamic efficiency. As suspected, the distance between the FSL and the wingtip vortex is greatly reduced in the 11-hole wing across all angles of attack when compared to the baseline. The wingtip vortex in the 11-hole wing case is much closer to the FSL when compared to the baseline. The proximity of the wingtip vortex to the FSL at all angles of attack could be the reason why the azimuthal velocity profiles deviated a lot from the Batchelor's model in Figure 16. The reduced FSL-wingtip vortex distance could be due to the reduction in downwash angle of air coming through the hole. According to thin airfoil theory, the downwash angle ϵ is directly proportional to the lift coefficient.

$$\epsilon = \frac{C_L}{\pi e AR} \tag{14}$$

where e is the span efficiency and AR is the aspect ratio. Therefore, from the lift coefficient variation (Figure 6b), it is expected that the downwash angle of the 11-hole wing will be lower when compared to the wingtip vortex at higher angles of attack. At a 8° angle of attack, the downwash angle for the baseline wing from Equation 13 is 4.5° and for the 11-hole wing, the downwash angle is 4°. Therefore there is a change of 12.5% in the downwash angle at 8° angle of attack which is enough to increase the FSL-wingtip vortex interaction. To quantify the reduction in the wingtip vortex strength, wingtip vortex circulation was determined by integrating the vorticity as shown in Equation 15.

$$\Gamma = \iint \omega . ds \tag{15}$$

where Γ is the circulation, ω is the vorticity and ds is the incremental surface area. After determining the wingtip vortex center through Q-criterion, several concentric circles were made around the vortex center to determine the circulation as a function of wingtip vortex radius. An example of a circulation plot obtained through this procedure is shown in Figure 18a. Once the circulation variation is obtained, the profile is fitted to an ideal Lamb-Oseen vortex model described by Equation 16 to determine the total circulation Γ_0 and vortex core radius r_c.

$$\Gamma(r) = \Gamma_0 \left(1 - \exp\left(-\frac{r^2}{r_c^2}\right)\right) \tag{16}$$

This method of finding circulation was developed by Stevens (2013) and has been employed in Corkery and Babinsky (2017) and Stevens and Babinsky (2017) to determine circulation of vortices emanating from pitching and plunging wings. The Lamb-Oseen model is fitted with the experimental data for the baseline NACA 0012 wing at 8° angle of attack by changing the total circulation value (Γ_0) (Figure 18b). A value of 0.3 m^2/s gave the highest R^2 value of 0.98. Similar fit was obtained for all angles of attack by changing the Γ_0 that gives a higher R^2 value for both the baseline and the 11-hole wing case. The average R^2 value for all the cases was 0.98.

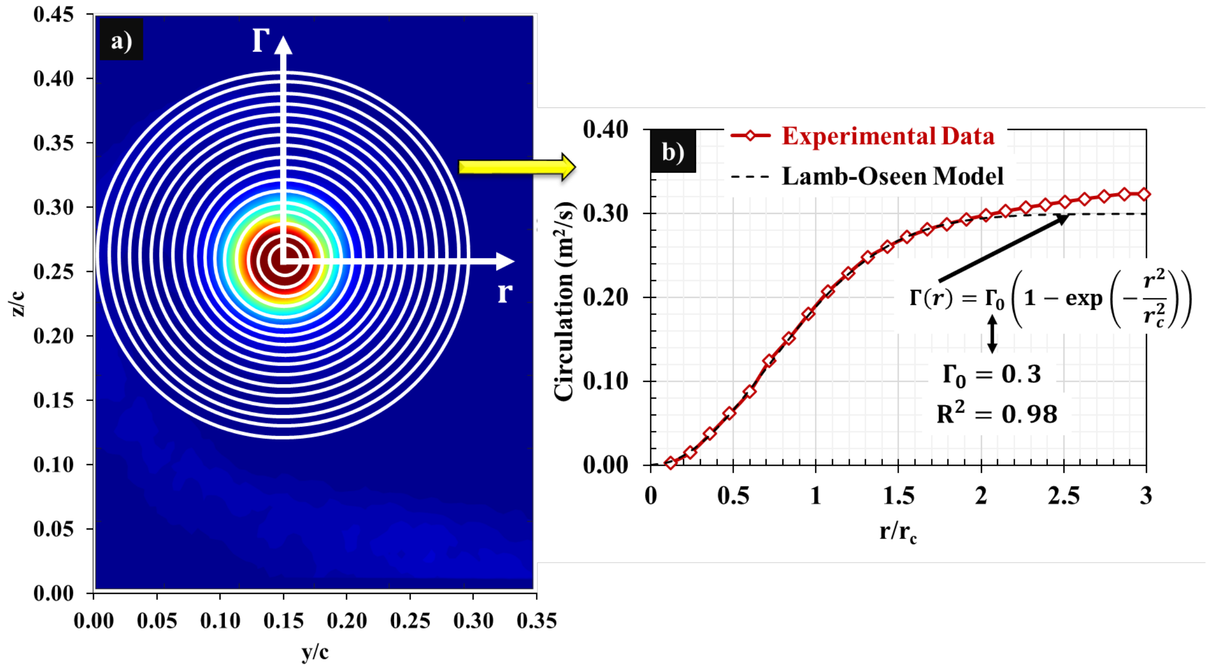

Figure 18 a) Variation of circulation as a function of vortex radius. The circulation was determined by taking concentric circles from the vortex center. b) Determination of total circulation of the wingtip vortex by fitting the experimental data with the Lamb-Oseen vortex model. The current graph is for the baseline NACA 0012 wing at 8° angle of attack.

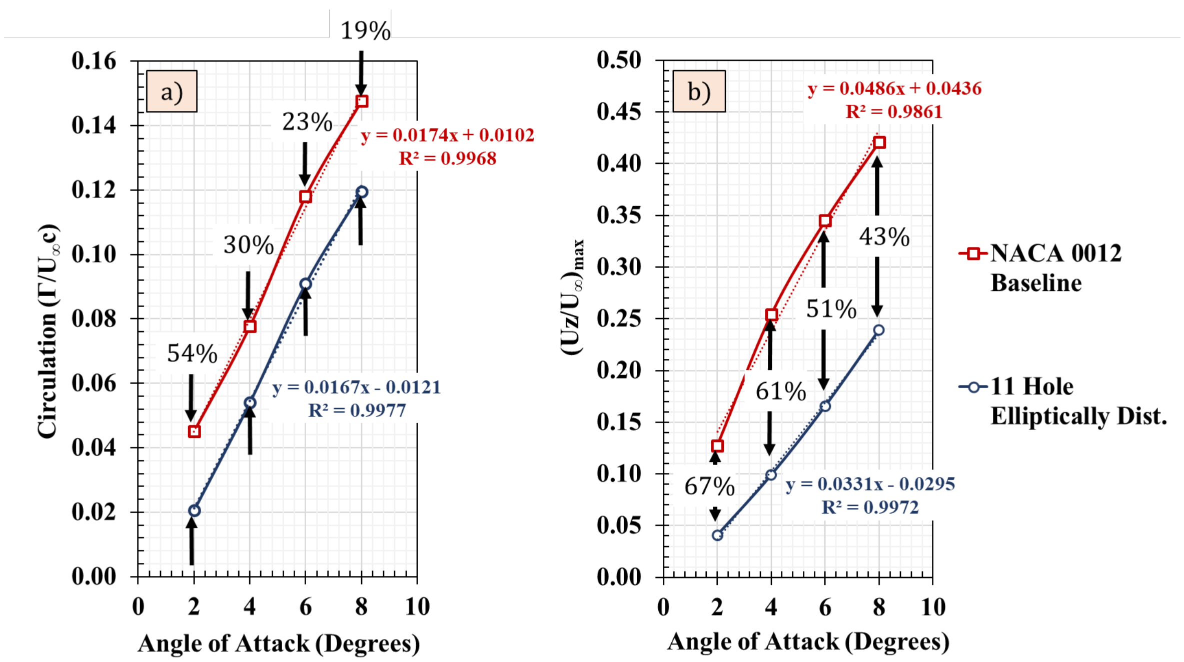

Figure 19 a) Variation of normalized circulation with the angle of attack for baseline wing and 11-hole wing. The circulation from the 11-hole wing is significantly lower than the baseline. b) The same trend is mirrored in the peak azimuthal velocity of the wingtip vortex.

The normalized circulation obtained by this method for all angles of attack is shown in Figure 19a for the NACA 0012 baseline wing and the 11-hole elliptically spaced wing. The circulation increases linearly (as denoted by the R^2 values) as a function of angle of attack for both cases. This is expected as the variation of the lift coefficient with angle of attack is also linear for both cases at the range of angle of attack shown.

The magnitude of the circulation however is significantly different. The 11-hole wing case circulation magnitude is significantly lower than the baseline wing at all angles of attack. The differences in the circulation however decreases with increase in angle of attack from 54% at 2° to 19% at 8° angle of attack. Similar trend is seen in the peak azimuthal velocity as a function of angle of attack shown again in Figure 19b. The average reduction in circulation across angles of attack however is around 30% between the baseline and the 11-hole wing. This result once again shows that added momentum in the free shear layer wake highly influenced the rollup of the wingtip vortex.

3.3.5 Wingtip Vortex RMS

The added streamwise momentum from the holes created an uneven x-vorticity distribution in the FSL when compared to the baseline case (Figure 16). The root-mean-square fluctuations in the FSL and the wingtip vortex was quantified to gain insight to this observation. The freestream normalized Y_{RMS} and Z_{RMS} is compared between the baseline and the 11-hole wing (Figure 20) at 6° and 8° angle of attack. The RMS was calculated using Equation 17.

$$Y_{RMS} = \sqrt{\overline{{u'_Y}^2}} \qquad\qquad Z_{RMS} = \sqrt{\overline{{u'_z}^2}} \tag{17}$$

where u'_Y is the fluctuating velocity about the Y-axis (parallel to the wingspan) and u'_z is the fluctuating velocity about the Z-axis (perpendicular to the wingspan). In the baseline case, both the Y_{RMS} and Z_{RMS} were concentrated around the vortex core. The normalized RMS values of $1.8 * 10^{-5}$and $2.0 * 10^{-5}$ are highlighted in white to distinctly observe the wingtip vortex and the FSL.

The RMS in the wingtip vortex core is higher when compared to the Z_{RMS} in the free shear layer. This indicates that the velocity fluctuations in Y and Z directions are greater at the wingtip vortex core when compared to the wingtip vortex boundary and the FSL. In the 11-hole wing case, however, the magnitude of Y_{RMS} and Z_{RMS} is distributed evenly through the vortex (except at the core) throughout the vortex when compared to the baseline. Similar Y_{RMS} is seen in the FSL and the wingtip vortex surrounding the core in the 11-hole wing case. A distinct separation can be made between the wingtip vortex and the FSL in the baseline case and shown by the wingtip vortex-FSL interface in Figure 20b. The Z_{RMS} at this interface is lower than the wingtip vortex and the FSL. There seems to be a disconnect between the Z_{RMS} between the FSL and in the wingtip vortex. However, in the 11-hole wing case, such interface is not observed. The turbulence in the FSL continues to roll-up into the wingtip vortex increasing the Z_{RMS} of the wingtip vortex surrounding the core.

The absence of the FSL-wingtip vortex interface in the 11-hole wing case further validates the interaction of the wingtip vortex and the FSL which lead to substantial reductions in peak azimuthal velocity and circulation. Since the azimuthal velocity is drastically decreased in the 11-hole wing case, it is likely that axial velocity component also decreased (according to Batchelor's model) making the wingtip vortex weaker when compared to the baseline. It was suspected that the vortex wandering may have influenced the FSL-wingtip vortex interface region in the 11-hole wing case. But as shown in Figure 11, both the baseline and the 11-hole wing case shows similar wandering of less than 5% of the chord. The wandering corrected Z_{RMS} at the vortex center only differed from the non-wandering corrected Z_{RMS} by less than 4%. But the differences seen between the baseline and the 11-hole wing are much greater. Similar results were observed with the 2° and 4° angle of attack cases as well.

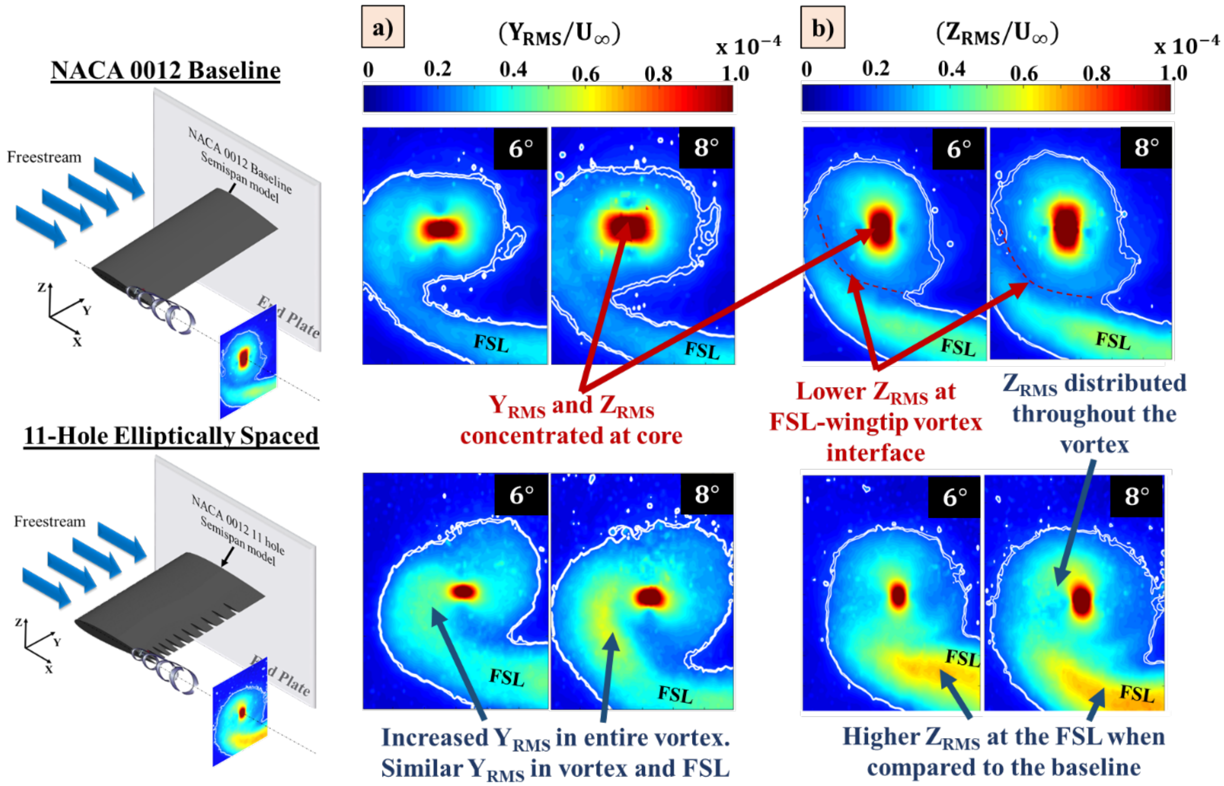

Figure 20 a) Normalized Y_{RMS} and b) Normalized Z_{RMS} variation across different angles of attack for baseline NACA 0012 wing and 11-hole elliptically spaced wing. Higher fluctuations are seen in the wingtip vortex and RMS in the 11-hole case when compared to the baseline.

3.3.6 Wingtip Vortex Mean Entropy Generation

The entropy generation is often associated with waste of energy, and in fluid dynamic systems, it can directly be translated into drag force by the Oswatitsch equation (1980) which states that the total drag force is directly proportional to the total increase in entropy.

$$DU_\infty = T_\infty \iint s\,\rho\,(\vec{U}.\vec{n})\,dA \tag{18}$$

where D represents the drag force, U_∞ is the freestream velocity, T_∞ is the freestream temperature, s is the entropy, ρ_∞ is the freestream density and $\vec{U}$ is the velocity vector. It is reasonable to infer from Equation 18 that wings which produce higher entropy produces higher drag. Therefore, the mean entropy generation in the wingtip vortex is determined for the baseline and the 11-hole wing case. Mean entropy generation in a fluid flow depends on the viscous dissipation of the mean flow (change of kinetic energy to internal energy) and the heat transfer as shown in Equation 19.

$$\dot{S}_{gen} = \frac{1}{T_0}\left(\tau_{ij}\frac{\partial U_i}{\partial x_j}\right) - \frac{1}{T_0^2}\left(q_j\frac{\partial T}{\partial x_j}\right) \tag{19}$$

where T_0 is the reference temperature, τ_{ij} is the viscous stress tensor, q_j is the rate of heat transfer and T is the temperature of the flow. By the definition of viscous dissipation and Fourier's law of a two dimensional flow in the YZ-plane, the first and second term in the right hand side of Equation 19 can be expanded as,

$$\frac{1}{T_0}\left(\tau_{ij}\frac{\partial U_i}{\partial x_j}\right) = \frac{\mu}{T_o}\left\{2\left(\frac{\partial U_Y}{\partial y}\right)^2 + 2\left(\frac{\partial U_Z}{\partial z}\right)^2 + \left(\frac{\partial U_Z}{\partial y} + \frac{\partial U_Y}{\partial z}\right)^2\right\} \tag{20}$$

$$\frac{1}{T_0^2}\left(q_j\frac{\partial T}{\partial x_j}\right) = \frac{k}{T_0^2}\left\{\left(\frac{\partial T}{\partial z}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2\right\} \tag{21}$$

The magnitude of the temperature gradient terms in the wingtip vortex can be considered negligible when compared to the magnitude of the velocity gradient terms. Therefore, the total entropy generation in the wingtip vortex can be estimated by,

$$\dot{S}_{gen} = \frac{\mu}{T_o}\left\{2\left(\frac{\partial U_Y}{\partial y}\right)^2 + 2\left(\frac{\partial U_Z}{\partial z}\right)^2 + \left(\frac{\partial U_Z}{\partial y} + \frac{\partial U_Y}{\partial z}\right)^2\right\} \tag{22}$$

The velocity gradients in Equation 22 was determined through central difference technique from the experimental data. The normalized entropy generation in the wingtip vortex for the baseline and the 11-hole wing is shown in Figure 21a for 6° and 8° angle of attack.

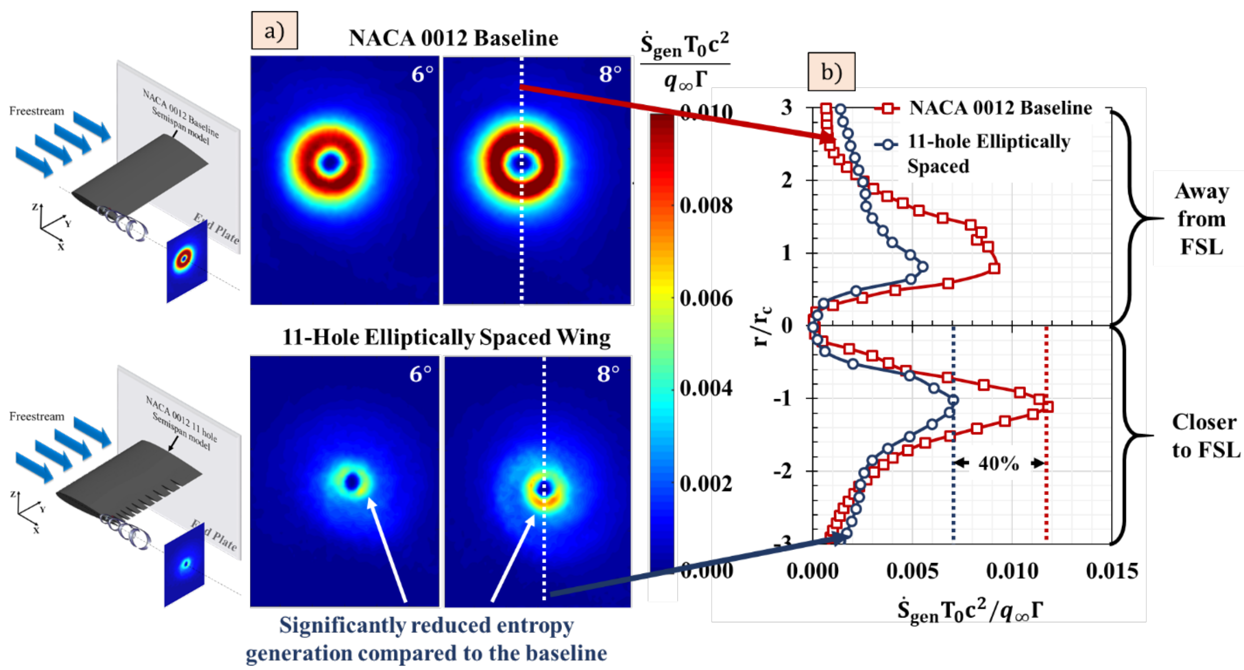

Figure 21 a) Normalized entropy generation contour indicating drastic differences in the wingtip vortex entropy generation between the baseline and 11-hole wing. b) Normalized entropy generation profiles at 8° angle of attack showing a peak entropy generation reduction of 40% in the 11-hole wing case when compared to the baseline.

The normalized entropy is defined as $\dot{S}_{gen}T_0c^2/q_\infty\Gamma$ where c is the chord of the wing, q_∞ is the dynamic pressure, Γ is the circulation of the wingtip vortex. The baseline case shows higher entropy generation in the wingtip vortex boundary where the velocity gradients are prominent. In the vortex boundary, the mean kinetic energy is being converted into the internal energy resulting in an entropy increase. But the 11-hole wing case shows a significant decrease in the entropy generation in the core boundary. This could be due to the reduction in the magnitude of the velocity gradients which in turn could be due to the increased interaction of the FSL and the wingtip vortex. Similar trends were observed at the 2° and 4° angle of attack. The normalized entropy generation profile for 8° angle of attack is plotted against the normalized radius in Figure 21b. The bimodal profile indicates an asymmetric entropy generation for both the cases. The lower part of the profile contains the FSL rollup into the vortex due to which it experiences higher entropy generation. The magnitude of the entropy generation is drastically reduced in the 11-hole case when compared to the baseline. The peak entropy generation reduced by 40% in the wingtip vortex due to the increased streamwise momentum further validating the effect of the holes in the wingtip vortex. It should be noted that the entropy generation shown in Figure 21 represents the mean entropy generation calculated from the velocity gradients. The mean entropy generation does not give any insight into the entropy generation through fluctuating quantities. The entropy generation from the fluctuating velocities will be discussed in the follow-up paper.

4. Conclusion

Three configurations of NACA 0012 semi-span wing model of AR 2 were investigated at the University of Dayton Low-Speed Wind Tunnel: 1) Eight holes equally spaced, 2) Eleven holes elliptically spaced (to mirror the lift distribution on the wing) and 3) No holes. The important conclusions from the research are:

1. The differences in the lift coefficient between the baseline, 8-hole and the 11-hole wing at lower angles of attack were almost negligible. The 11-hole wing had lower lift coefficient by 12% and lower induced drag by 20% when compared to the baseline at higher angles of attack. The total drag coefficient of the all three configurations only varied within 4%.
2. Achieving similar aerodynamic performance with a weight reduction of 13% and wingtip vortex circulation reduction by 30% in the 11-hole elliptically spaced wing is desirable in applications such as airplane wings, wind turbines, propellers, helicopter rotors, etc.
3. The reduced downwash angle in the 11-hole wing case increased the FSL-wingtip vortex interaction reducing the azimuthal velocity by 50% and the circulation by 30% when compared to the baseline. Implications of this result is that with a weaker vortex, the landing times between airplanes at the airport can be reduced which increases airport capacity.
4. The increased RMS values in the 11-hole wing case at the FSL-wingtip vortex interface further validated the interaction of the wingtip vortex and FSL which lead to significant reductions in wingtip vortex properties. Higher fluctuations in the wingtip vortex results in increased turbulent dissipation which also has implications on reducing landing times between airplanes at the airport.
5. The mean entropy generation was reduced by 40% in the 11-hole wing case when compared to the baseline.

The results show a strong evidence that the wingtip vortex was severely affected by the streamwise momentum of air coming through the wing at the wingtip which has several implications on variety of aerodynamic applications.

5. Future Work:

Work is currently being done to determine the streamwise and cross-stream velocity component spectrum at the wingtip vortex core and the boundary between the 11-hole wing and the baseline case through hot-wire anemometry. The results from hot-wire anemometry, the entropy generation through fluctuating quantities along with sensitivity analysis on the diameter of the holes will be presented in a follow-up paper.

6. Acknowledgements:

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